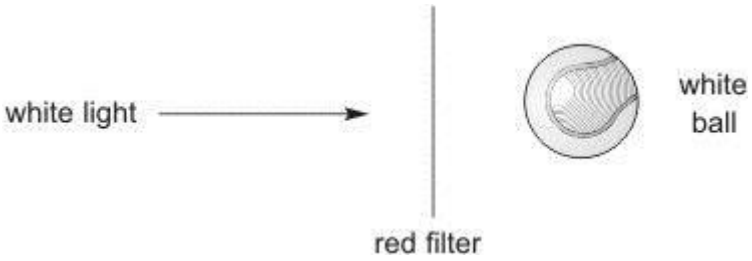
Q1.

(a) Peter had two different coloured tennis balls as shown below.



He shone white light through a red filter onto each ball.

(i) **experiment 1**



The white ball appeared red.
Explain why this ball appeared red.

.....

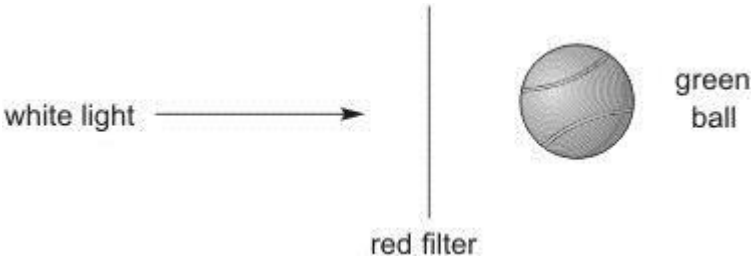
.....

.....

.....

2 marks

(ii) **experiment 2**



What colour did this ball appear?

.....

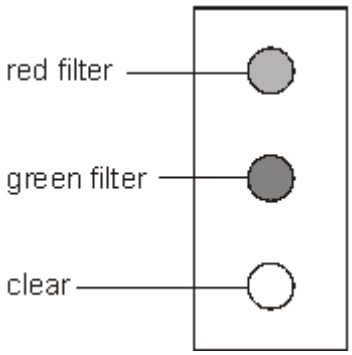
Explain your answer.

.....

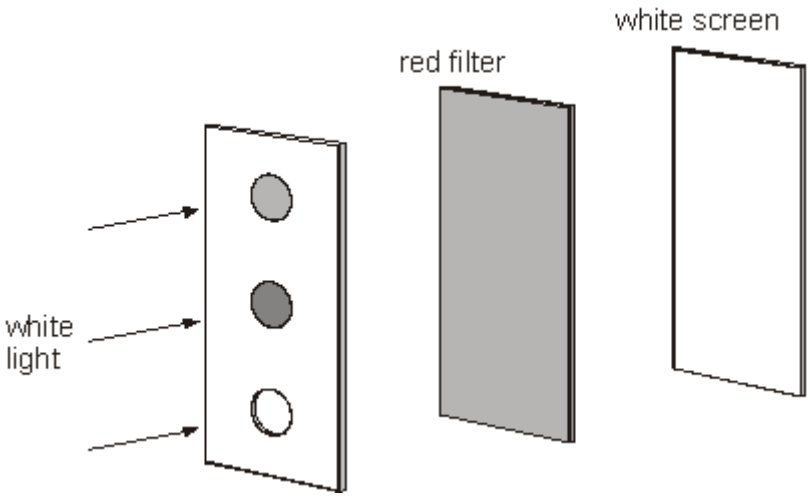
.....

2 marks

(b) Peter set up a different experiment.
He cut three holes in a piece of card.
Two of the holes were covered by coloured filters as shown below.



Peter placed a red filter between the piece of card and a white screen.
He shone white light at the piece of card with three holes in it.



What would Peter see on the screen?

.....

.....

1 mark
maximum 5 marks

Q1.

(a) (i) • only red light passes through the filter

*accept 'the other colours are filtered out **or** absorbed'*

'red light passes through' is insufficient

*a mark for this answer may be awarded **in either a i or a ii** provided there is no contradiction*

1 (L7)

• the ball reflects red light

accept 'white objects reflect all colours'

1 (L7)

(ii) black

accept 'you cannot see it'

1 (L7)

any **one** from

• the green ball does not reflect red light or the light that passes through the filter

• the ball absorbs red light

accept 'no green light reaches the ball'

1 (L7)

(b) two red spots

accept 'red black red'

'two spots' is insufficient

'red spots' is insufficient

*do **not** accept 'only red light'*

1 (L7)

[5]